

Below is an expanded explanation guide you can use while presenting.

Hardware and Operating System Fundamentals

Expanded Instructor Guide with Full Forms and Real-World Examples

1. Basic Terms

Hardware

Definition: Hardware means the physical parts of a computer that can be seen or touched.

Examples

- Keyboard
- Monitor
- Processor
- Memory
- Hard disk
- Network card
- Motherboard

Real-world example

When a laptop screen breaks, it is a hardware problem. When Microsoft Word stops responding, it is usually a software problem.

Software

Definition: Software is a set of instructions that tells the computer hardware what to do.

Examples

- Microsoft Word
- Google Chrome
- Antivirus software
- Windows
- Linux
- Calculator application

Real-world example

Google Chrome is stored on the disk as software. When you open Chrome, the operating system loads it into memory and the CPU begins executing its instructions.

OS: Operating System

Full form: Operating System

Definition: An operating system is the main software that manages the computer's hardware, applications, users, files, memory, and security permissions.

Examples

- Microsoft Windows
- Linux
- macOS
- Android
- iOS

Real-world example

When you connect a USB drive, the operating system:

1. Detects the USB device.
2. Loads the correct driver.
3. Assigns it a drive location.
4. Reads its file system.
5. Checks whether the user has permission.
6. Displays its contents.

Cybersecurity connection

If an attacker compromises one application, the damage may be limited. If the attacker gains control of the operating system, they may access files, users, processes, network connections, and security settings.

2. CPU

CPU: Central Processing Unit

Full form: Central Processing Unit

Definition: The CPU is the component that executes instructions and performs calculations.

It is often called the **brain of the computer**, although it works together with memory, storage, and the operating system.

Real-world example

When you enter a password:

1. The keyboard sends the characters to the computer.
2. The password is temporarily stored in memory.
3. The CPU processes the password.
4. The operating system or application compares it with the expected authentication information.
5. Access is allowed or denied.

ALU: Arithmetic Logic Unit

Full form: Arithmetic Logic Unit

Definition: The ALU is the part of the CPU that performs mathematical and logical operations.

Operations

- Addition
- Subtraction
- Comparison
- AND operations
- OR operations
- Greater-than and less-than checks

Real-world example

When an application checks:

Is the user's age greater than or equal to 18?

The CPU uses logical comparison operations to determine whether the condition is true or false.

CU: Control Unit

Full form: Control Unit

Definition: The Control Unit coordinates the CPU's activities. It fetches instructions, understands them, and directs other parts of the processor.

Real-world analogy

The Control Unit acts like a classroom instructor:

- It gives instructions.
- It decides the sequence of activities.
- It coordinates different students and resources.

Register

Definition: A register is a very small and extremely fast storage location inside the CPU.

Registers hold information that the CPU is currently using.

Real-world example

During a calculation such as:

$$25 + 15 = 40$$

The CPU may temporarily place the values 25, 15, and the result 40 in registers.

Cybersecurity connection

Some malware and exploit techniques manipulate register values to redirect program execution.

Cache

Definition: Cache is a small, fast memory located inside or very close to the CPU. It stores frequently used instructions and data.

Common cache levels

- **L1:** Level 1 cache, smallest and fastest
- **L2:** Level 2 cache, larger but slightly slower
- **L3:** Level 3 cache, larger and often shared between CPU cores

Real-world analogy

- Storage is a library.
- RAM is your study table.
- CPU cache is the notebook beside your hand.
- Registers are the information you are currently thinking about.

Cybersecurity connection

Some side-channel attacks study differences in cache access time to infer sensitive information.

Core

Definition: A core is an individual processing unit inside a CPU.

Example

- Dual-core means two processing cores.
- Quad-core means four processing cores.
- Octa-core means eight processing cores.

Real-world example

One core may process a browser task while another processes an antivirus scan. The operating system decides how the work is distributed.

GHz: Gigahertz

Full form: Gigahertz

Definition: Gigahertz measures the number of CPU clock cycles per second.

1 GHz = approximately 1 billion cycles per second

A 3.2 GHz processor has a clock operating at approximately 3.2 billion cycles per second.

Important clarification

A higher GHz value does not always mean the entire computer will be faster. Performance also depends on:

- Number of cores
- CPU architecture
- Cache size
- RAM speed
- Storage speed
- Application design
- System temperature

Ring 0 and Ring 3

Ring 0

Definition: Ring 0 is the most privileged CPU operating level. The operating-system kernel normally operates here.

Ring 3

Definition: Ring 3 is a restricted operating level used by normal applications.

Real-world analogy

Think of an airport:

- **Ring 3:** Passenger waiting area
- **Ring 0:** Air-traffic control center

Passengers can use airport services, but they cannot enter the control center.

Cybersecurity example

If malicious code runs only inside a normal application, its actions may be restricted. If it reaches Ring 0, it may:

- Modify the operating system
- Access protected memory
- Disable security tools
- Control devices
- Hide malicious activity

Privilege Escalation

Definition: Privilege escalation is the process of gaining permissions that a user or program was not originally authorized to have.

Example

An attacker initially gains access as a normal user. They exploit a vulnerability and become an administrator.

Two common types

- **Vertical privilege escalation:** Moving from a normal user to administrator or root
- **Horizontal privilege escalation:** Accessing another user's data without gaining a higher role

Real-world example

A student should only access their own examination result. If the student changes a number in a website address and views another student's result, that is horizontal privilege escalation.

Side-Channel Attack

Definition: A side-channel attack obtains information by observing indirect system behavior instead of directly breaking the security mechanism.

Information attackers may observe

- Processing time
- Power consumption
- CPU cache behavior
- Electromagnetic signals
- Sound
- Temperature changes

Real-world analogy

You cannot see what a person is cooking, but you observe the smell, heat, time, and sounds. From these indirect clues, you try to identify the food.

3. RAM and Memory

RAM: Random Access Memory

Full form: Random Access Memory

Definition: RAM is temporary, high-speed memory used for programs and data that are currently active.

Important property

RAM is normally **volatile**, which means its contents are lost when electrical power is removed.

Real-world example

When you open Microsoft PowerPoint:

1. PowerPoint is stored on the SSD.
2. The OS loads required PowerPoint files into RAM.
3. The CPU executes PowerPoint instructions.
4. The presentation you are editing is processed in RAM.

5. When you save, the information is written to storage.

Classroom analogy

- Storage is the school cupboard.
- RAM is the student's desk.
- CPU is the student.
- A file is taken from the cupboard and placed on the desk before being used.

Volatile Memory

Definition: Volatile memory requires power to retain information.

Example

Unsaved work may disappear when:

- The computer suddenly loses power
- The battery becomes completely empty
- The operating system crashes
- The application closes unexpectedly

Virtual Memory

Definition: Virtual memory is a memory-management system that gives each process its own logical address space and can use storage to support RAM.

Real-world example

If RAM is heavily used, the OS may move less frequently used memory information to a page file or swap area on storage.

Important clarification

Virtual memory is useful, but storage is much slower than physical RAM. Heavy paging can make a computer feel slow.

Page File

Definition: A page file is an area of storage used by Windows to support virtual memory.

Common Windows filename:

pagefile.sys

Swap

Definition: Swap is storage space used by Linux and Unix-like systems to support memory management.

It may be implemented as:

- A swap partition
- A swap file

RAM Scraping

Definition: RAM scraping is the process of searching memory for sensitive information.

Possible targets

- Passwords
- Payment-card information
- Authentication tokens
- Encryption keys
- Personal data

Real-world example

Some payment-system malware has searched active memory for card information while transactions were being processed.

Fileless Malware

Definition: Fileless malware operates mainly in memory and may use legitimate system tools instead of relying only on a traditional malicious executable file.

Commonly abused tools

- PowerShell
- Windows Management Instrumentation
- Command shells
- Script interpreters
- Office macros

WMI: Windows Management Instrumentation

Full form: Windows Management Instrumentation

Definition: WMI is a Windows technology used to manage and obtain information about computers.

Important clarification

PowerShell and WMI are legitimate administration tools. Their presence does not mean a system is infected. Attackers may misuse them because they are normally trusted and already available.

Memory Forensics

Definition: Memory forensics is the examination of information captured from RAM during a cybersecurity investigation.

Investigators may find

- Running processes
- Network connections
- Injected code
- Command history
- Encryption material
- Suspicious drivers
- Malware that did not create an obvious disk file

Cold Boot Attack

Definition: A cold boot attack attempts to recover information that remains briefly in memory after a computer is restarted or loses power.

Security controls

- Physical security
- Fully shutting down sensitive devices
- Modern memory-protection technologies
- Secure hardware configuration
- Preventing unauthorized physical access

4. Storage

HDD: Hard Disk Drive

Full form: Hard Disk Drive

Definition: An HDD is a storage device that uses rotating magnetic disks and mechanical components.

Advantages

- Lower cost for large storage capacities
- Suitable for backups and bulk data

Disadvantages

- Slower than an SSD
- Contains moving parts
- More vulnerable to physical shock

SSD: Solid-State Drive

Full form: Solid-State Drive

Definition: An SSD stores information using flash memory and has no moving mechanical parts.

Advantages

- Faster startup
- Faster application loading
- Lower mechanical-failure risk
- Quiet operation

Real-world example

A laptop with an SSD may start Windows and open applications faster than a similar laptop using an HDD.

Non-Volatile Storage

Definition: Non-volatile storage retains data when power is removed.

Examples

- HDD
- SSD
- USB flash drive
- Memory card

USB: Universal Serial Bus

Full form: Universal Serial Bus

Definition: USB is a standard used to connect devices such as keyboards, storage drives, mobile phones, and security keys.

Cybersecurity example

An unknown USB drive may contain:

- Malicious documents
- Malware
- A fake keyboard-style device
- Files designed to exploit an unpatched system

Safe teaching point

“Do not connect unknown USB devices to production or personal computers.”

Data at Rest

Definition: Data at rest means information stored on a device and not currently moving through a network.

Examples

- Files on an SSD
- Database records
- Backup files
- Information on a USB drive

Encryption

Definition: Encryption converts readable information into an unreadable form using an algorithm and a key.

Real-world example

If an encrypted laptop is stolen, the thief should not be able to read the stored files without the required key or authentication.

BitLocker

Definition: BitLocker is Microsoft’s full-volume encryption technology for supported Windows editions.

Cybersecurity purpose

It helps protect stored information if a computer or disk is lost, stolen, or removed.

Important clarification

If the computer is already unlocked and malware is running as an authorized user, the malware may access files that are available to that user.

Persistence

Definition: Persistence is an attacker's ability to maintain access or restart malicious activity after a reboot, logout, or interruption.

Common persistence methods

- Startup applications
- Scheduled tasks
- Services
- Registry entries
- Browser extensions
- Modified scripts
- Compromised user accounts

Real-world example

Malware creates a scheduled task that starts every time the user signs in.

Ransomware

Definition: Ransomware is malware designed to deny access to data or systems, commonly by encrypting files and demanding payment.

Common impact

- Business interruption
- Data loss
- Recovery expenses
- Reputational damage
- Possible data exposure

Defenses

- Offline or protected backups
- Endpoint protection
- Software updates

- Least privilege
- Email filtering
- Network segmentation
- User awareness

5. Firmware and Booting

BIOS: Basic Input/Output System

Full form: Basic Input/Output System

Definition: BIOS is older firmware that initializes computer hardware and starts the operating-system boot process.

UEFI: Unified Extensible Firmware Interface

Full form: Unified Extensible Firmware Interface

Definition: UEFI is a modern firmware interface that replaces many functions of the traditional BIOS.

UEFI benefits

- Support for modern storage
- Improved startup capabilities
- Secure Boot support
- More flexible configuration

Firmware

Definition: Firmware is software stored in a hardware device that controls its low-level operation.

Examples

- Laptop UEFI
- Router firmware
- SSD firmware
- Keyboard firmware
- Printer firmware

Cybersecurity example

A router with outdated firmware may contain a vulnerability that allows unauthorized access.

POST: Power-On Self-Test

Full form: Power-On Self-Test

Definition: POST is the initial check performed during computer startup to verify that essential hardware is available and functioning.

Possible checks

- CPU
- RAM
- Keyboard
- Graphics hardware
- Storage devices

Real-world example

A pattern of beeps during startup may indicate a hardware problem detected during POST.

Bootloader

Definition: A bootloader is a small program that locates and starts loading the operating-system kernel.

Examples

- Windows Boot Manager
- GRUB on many Linux systems

GRUB: GNU GRand Unified Bootloader

Full form: GNU GRand Unified Bootloader

Definition: GRUB is a commonly used bootloader for Linux and other operating systems.

Secure Boot

Definition: Secure Boot is a UEFI security feature that helps prevent unauthorized or untrusted software from loading during the startup process.

Real-world example

If an attacker replaces a trusted boot component with unauthorized code, Secure Boot may prevent it from starting if its signature is not trusted.

TPM: Trusted Platform Module

Full form: Trusted Platform Module

Definition: A TPM is a security component used to protect cryptographic information and support system-integrity checks.

Common uses

- BitLocker key protection
- Device authentication
- Secure Boot measurements
- Windows Hello

6. Networking

NIC: Network Interface Card

Full form: Network Interface Card

Definition: A NIC allows a computer to connect to and communicate over a network.

Types

- Ethernet NIC
- Wi-Fi adapter
- Virtual network adapter
- Fiber network adapter

MAC Address

Full form: Media Access Control address

Definition: A MAC address identifies a network interface at the local-network or data-link layer.

Example format

00-1A-2B-3C-4D-5E

Important clarification

MAC addresses are mainly relevant to local-network communication and can be changed or spoofed in many environments.

IP Address

Full form: Internet Protocol address

Definition: An IP address identifies a device or network interface for communication over an IP network.

Example IPv4 address

192.168.1.25

IPv4: Internet Protocol version 4

Full form: Internet Protocol version 4

IPv6: Internet Protocol version 6

Full form: Internet Protocol version 6

Example IPv6 address

2001:db8::10

LAN: Local Area Network

Full form: Local Area Network

Definition: A LAN connects devices within a limited area, such as a home, school, laboratory, or office.

WAN: Wide Area Network

Full form: Wide Area Network

Definition: A WAN connects devices or networks across larger geographical areas.

The internet is the largest commonly known WAN.

ARP: Address Resolution Protocol

Full form: Address Resolution Protocol

Definition: ARP helps an IPv4 device identify the MAC address associated with another local IPv4 address.

Real-world analogy

You know a student's name, but you need to know which classroom seat belongs to that student.

ARP Spoofing

Definition: ARP spoofing occurs when a system sends false ARP information to misdirect local-network traffic.

Possible impact

An attacker may attempt to:

- Intercept traffic
- Redirect traffic
- Disrupt communication
- Position themselves between two devices

MAC Spoofing

Definition: MAC spoofing means changing or imitating the MAC address used by a network interface.

Real-world example

A network may allow only registered device addresses. An unauthorized user may attempt to copy an approved device's MAC address.

Important clarification

MAC filtering alone is not strong authentication because MAC addresses can often be observed and changed.

Packet

Definition: A packet is a structured unit of data transmitted through a network.

Real-world analogy

A large parcel is divided into several smaller packages. Each package contains addressing information and part of the content.

Packet Sniffing

Definition: Packet sniffing is the process of capturing and examining network traffic.

Legitimate uses

- Troubleshooting
- Network monitoring
- Incident investigation
- Performance analysis

Malicious use

An attacker may attempt to collect unencrypted information or observe communication patterns.

7. Kernel and Processes

Kernel

Definition: The kernel is the central part of an operating system. It manages CPU usage, memory, devices, processes, and system calls.

Real-world analogy

The kernel is the central control room of the computer.

User Mode

Definition: User mode is a restricted operating environment used by normal applications.

Examples include:

- Browser
- Calculator
- PowerPoint
- Media player

Kernel Mode

Definition: Kernel mode is a highly privileged environment used by the operating-system kernel and trusted low-level components.

Cybersecurity importance

A kernel-mode compromise may allow an attacker to:

- Access protected memory
- Manipulate processes
- Control hardware
- Hide malicious activity
- Interfere with security tools

System Call

Definition: A system call is a controlled request from an application to the operating-system kernel.

Real-world example

When PowerPoint saves a presentation:

1. PowerPoint requests a file operation.
2. The request reaches the kernel.
3. The kernel checks permissions.
4. The file system handles the location.
5. A storage driver communicates with the SSD.
6. The result returns to PowerPoint.

API: Application Programming Interface

Full form: Application Programming Interface

Definition: An API is a defined way for software components to communicate or request services.

Real-world analogy

A restaurant menu is like an API. It shows customers what they can request without allowing them into the kitchen.

Process

Definition: A process is a running instance of a program.

Example

chrome.exe stored on a disk is a program. When Chrome is opened, one or more Chrome processes begin running.

PID: Process Identifier

Full form: Process Identifier

Definition: A PID is a number assigned by the operating system to identify a running process.

Example

chrome.exe PID 4820

PPID: Parent Process Identifier

Full form: Parent Process Identifier

Definition: A PPID identifies the process that started another process.

Security example

If Microsoft Word unexpectedly starts PowerShell, analysts may investigate:

- The document that was opened
- The PowerShell command
- The user account
- The file's source
- Connected network addresses

This behavior can sometimes be legitimate, so context is required.

Thread

Definition: A thread is a sequence of instructions executed within a process.

A process may contain one thread or many threads.

Analogy

- Process: A restaurant
- Threads: Workers performing different jobs inside it

Process Injection

Definition: Process injection is a technique in which code is placed or executed inside another running process.

Real-world analogy

An intruder enters a company while wearing the identity badge of a trusted employee.

Security impact

Injected code may inherit the appearance or access level of the targeted process.

Process Hollowing

Definition: Process hollowing is a technique where a legitimate process is started and its intended code is replaced or modified with different code.

Analogy

A genuine delivery vehicle enters the facility, but its original cargo is removed and replaced with unauthorized material.

8. File Systems and Permissions

File System

Definition: A file system is the method an operating system uses to name, store, organize, retrieve, and protect files.

NTFS: New Technology File System

Full form: New Technology File System

Definition: NTFS is a common Windows file system that supports permissions, encryption-related capabilities, large files, and auditing features.

FAT32: File Allocation Table 32

Full form: File Allocation Table 32

Definition: FAT32 is an older file system commonly used for compatibility with removable devices.

Limitation

FAT32 does not support individual files larger than approximately 4 GB.

exFAT: Extensible File Allocation Table

Full form: Extensible File Allocation Table

Definition: exFAT is a file system often used on large removable storage devices and supports files larger than the FAT32 limit.

ext4: Fourth Extended File System

Full form: Fourth Extended File System

Definition: ext4 is a commonly used Linux file system.

ACL: Access Control List

Full form: Access Control List

Definition: An ACL records which users or groups can access a resource and what actions they are allowed to perform.

Common permissions

- Read
- Write
- Execute

- Modify
- Delete
- Full control

Least Privilege

Definition: Least privilege means giving a user, process, or application only the permissions required to complete its authorized task.

Real-world example

A finance intern may need permission to read expense records but should not automatically receive permission to:

- Delete all records
- Create administrator accounts
- Disable security tools
- Modify operating-system files

Root

Definition: Root is the highest-privileged administrative account on Linux and Unix-like systems.

Administrator

Definition: An administrator is a user or role with elevated permission to manage a computer or environment.

Important practice

Administrators should use normal user accounts for ordinary work and elevate privileges only when required.

chmod

Meaning: Change mode

Definition: chmod is a Linux command used to modify file and directory permissions.

Permission values

- 4 means read
- 2 means write
- 1 means execute

Therefore:

- 7 means read + write + execute
- 6 means read + write
- 5 means read + execute

Example

```
chmod 777 example.sh
```

This gives read, write, and execute permissions to:

1. Owner
2. Group
3. Others

Security concern

Giving everyone full permission may allow an unauthorized user to change or replace the file.

9. Services, Protocols, and Ports

Service

Definition: A service is a program that normally operates in the background and provides a function to the system or network.

Examples

- Web server
- Database service
- DNS service
- Remote access service
- Update service

Port

Definition: A port is a logical number used to direct network communication to the correct service.

Analogy

- IP address: Office building address
- Port number: Room number

- Service: Team working in that room

SSH: Secure Shell

Full form: Secure Shell

Default port: TCP 22

Purpose: Secure remote login and command execution.

Real-world example

A Linux administrator uses SSH to manage a remote server.

DNS: Domain Name System

Full form: Domain Name System

Default port: UDP or TCP 53

Purpose: Translates domain names into IP addresses.

Example

example.com → IP address

Analogy

DNS is similar to a contact list that converts a person's name into a telephone number.

HTTP: Hypertext Transfer Protocol

Full form: Hypertext Transfer Protocol

Default port: TCP 80

Purpose: Transfers web content without providing transport encryption by itself.

HTTPS: Hypertext Transfer Protocol Secure

Full form: Hypertext Transfer Protocol Secure

Default port: TCP 443

Purpose: Transfers web content using encrypted connections, normally through TLS.

TLS: Transport Layer Security

Full form: Transport Layer Security

Definition: TLS helps provide encryption, integrity, and server authentication for network communication.

Real-world example

When visiting a legitimate HTTPS banking website, TLS helps prevent observers on the network from easily reading or modifying the session.

RDP: Remote Desktop Protocol

Full form: Remote Desktop Protocol

Default port: TCP or UDP 3389

Purpose: Allows users to access and control a Windows desktop remotely.

Cybersecurity risk

An internet-exposed RDP service may be targeted through:

- Password guessing
- Stolen credentials
- Unpatched vulnerabilities

Controls

- MFA
- Network restrictions
- VPN access
- Account lockout controls
- Strong passwords
- Security updates

MFA: Multi-Factor Authentication

Full form: Multi-Factor Authentication

Definition: MFA requires at least two different types of authentication evidence.

Factor types

- Something you know: Password or PIN
- Something you have: Phone or security key
- Something you are: Fingerprint or facial recognition

VPN: Virtual Private Network

Full form: Virtual Private Network

Definition: A VPN creates a protected communication connection between a device and another network or service.

MySQL

Meaning: MySQL is the proper name of a relational database-management system. “SQL” means Structured Query Language.

Common MySQL port: TCP 3306

TCP: Transmission Control Protocol

Full form: Transmission Control Protocol

Definition: TCP provides connection-oriented, ordered, and reliable data delivery.

UDP: User Datagram Protocol

Full form: User Datagram Protocol

Definition: UDP provides fast, connectionless communication without the same delivery guarantees as TCP.

Simple comparison

- TCP is like registered courier delivery.
- UDP is like quickly dropping messages into mailboxes without waiting for confirmation.

10. Memory-Security Protections

ASLR: Address Space Layout Randomization

Full form: Address Space Layout Randomization

Definition: ASLR randomizes where important program components are placed in memory.

Analogy

A valuable item is stored in a different locker every time. An attacker cannot rely on the same locker number.

DEP: Data Execution Prevention

Full form: Data Execution Prevention

Definition: DEP helps prevent code from running in memory regions intended only for data.

NX: No Execute

Full form: No Execute

Definition: NX is a hardware-supported protection that marks certain memory pages as non-executable.

Analogy

A page containing customer information should be treated as data, not as an instruction telling an employee what to do.

Stack

Definition: The stack is a region of memory used for temporary information related to function calls.

It may contain:

- Local variables
- Return addresses
- Function parameters
- Saved processor information

Buffer Overflow

Definition: A buffer overflow occurs when a program writes more information into a memory area than it was designed to hold.

Analogy

A glass is designed to hold 250 ml, but someone pours in 500 ml. The excess spills into nearby areas.

Cybersecurity impact

The overflow may corrupt nearby data and potentially influence program execution.

Stack Canary

Definition: A stack canary is a special value placed near sensitive stack information to help detect memory corruption.

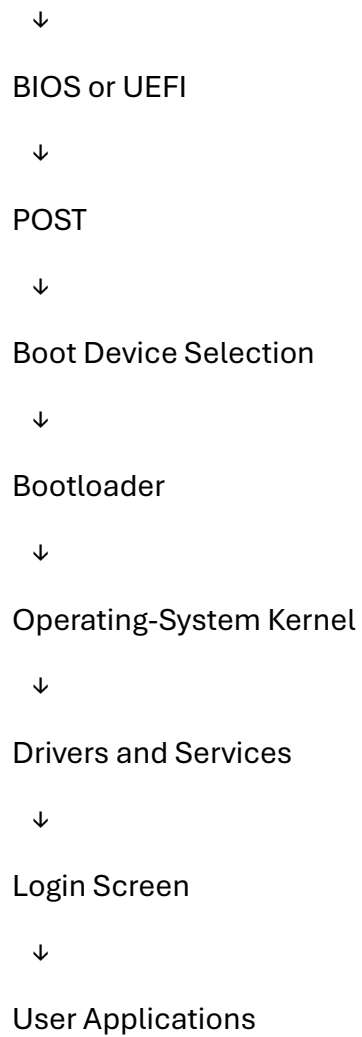
Analogy

A security seal is placed on a package. If the seal changes, the system knows someone may have tampered with the package.

11. Boot Process with Security Connections

Use this sequence on the board:

Power On



Driver

Definition: A driver is software that allows the operating system to communicate with a hardware device.

Examples

- Printer driver
- Graphics driver
- Network driver
- Storage driver

Cybersecurity connection

Drivers often operate with high privileges. A vulnerable or malicious driver may create serious security risk.

Bootkit

Definition: A bootkit is malicious software designed to interfere with or persist within the startup process.

Why it is dangerous

It may begin running before common user-level security applications.

Rootkit

Definition: A rootkit is a collection of techniques or software designed to maintain privileged access and hide malicious activity.

Possible capabilities

- Hide files
- Hide processes
- Hide network connections
- Manipulate system information
- Maintain privileged access

12. Complete Real-World Connection Example

Scenario: Student Opens a Website

1. **UEFI** previously initialized the hardware during startup.
2. The **bootloader** loaded the operating-system **kernel**.
3. The student opens a browser.
4. The browser becomes a **process** with a **PID**.
5. The OS loads required browser components from the **SSD** into **RAM**.
6. The **CPU** executes the browser instructions.
7. The browser asks the OS to connect to a website through an **API** and **system calls**.
8. **DNS** converts the website name into an **IP address**.
9. The **NIC** sends network packets.
10. **HTTPS** and **TLS** protect the web connection.
11. The website returns content.
12. The CPU processes the content and the graphics driver displays it.

Where an attacker may interfere

- Malicious browser extension
- Fake DNS response
- Unsecured Wi-Fi
- Browser vulnerability
- Stolen authentication token
- Malicious downloaded file
- Privilege-escalation vulnerability
- Weak file permissions

Defensive controls

- Browser and OS updates
- HTTPS
- Endpoint protection
- DNS security
- Least privilege
- MFA
- Safe download practices
- Network monitoring

Short Forms Quick Revision List

- **ACL:** Access Control List
- **ALU:** Arithmetic Logic Unit
- **API:** Application Programming Interface
- **ARP:** Address Resolution Protocol
- **ASLR:** Address Space Layout Randomization
- **BIOS:** Basic Input/Output System
- **CPU:** Central Processing Unit
- **CU:** Control Unit
- **DEP:** Data Execution Prevention
- **DNS:** Domain Name System

- **FAT32:** File Allocation Table 32
- **GHz:** Gigahertz
- **HDD:** Hard Disk Drive
- **HTTP:** Hypertext Transfer Protocol
- **HTTPS:** Hypertext Transfer Protocol Secure
- **IP:** Internet Protocol
- **LAN:** Local Area Network
- **MAC:** Media Access Control
- **MFA:** Multi-Factor Authentication
- **NIC:** Network Interface Card
- **NTFS:** New Technology File System
- **NX:** No Execute
- **OS:** Operating System
- **PID:** Process Identifier
- **POST:** Power-On Self-Test
- **PPID:** Parent Process Identifier
- **RAM:** Random Access Memory
- **RDP:** Remote Desktop Protocol
- **SSD:** Solid-State Drive
- **SSH:** Secure Shell
- **TCP:** Transmission Control Protocol
- **TLS:** Transport Layer Security
- **TPM:** Trusted Platform Module
- **UDP:** User Datagram Protocol
- **UEFI:** Unified Extensible Firmware Interface
- **USB:** Universal Serial Bus
- **VPN:** Virtual Private Network
- **WAN:** Wide Area Network

- **WMI:** Windows Management Instrumentation

Teaching tip

Whenever a short form first appears, say the full form immediately:

“CPU means Central Processing Unit. It executes the computer’s instructions.”